Mobile Development :

2 : Mobile Apps : Idea, Planning, Design and Development

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Outline :

- App Development Process
  1. *Idea & Market Research*
  2. *Planning and Strategy*
  3. *UX and Design*
  4. *Development*
  5. *Testing*
  6. *Publishing and Maintenance*
- Usability & UX : Design Principles
- Design Patterns for Mobile Apps
- Business Models for Apps

In this course :

You will learn how to build an App ?

In this course :

Or, You will learn how to build a Business with an App ?

Warning :

Google and Apple will remove Apps from their stores which they don't meet their requirements, making it difficult for poorly planned and executed apps to stay online

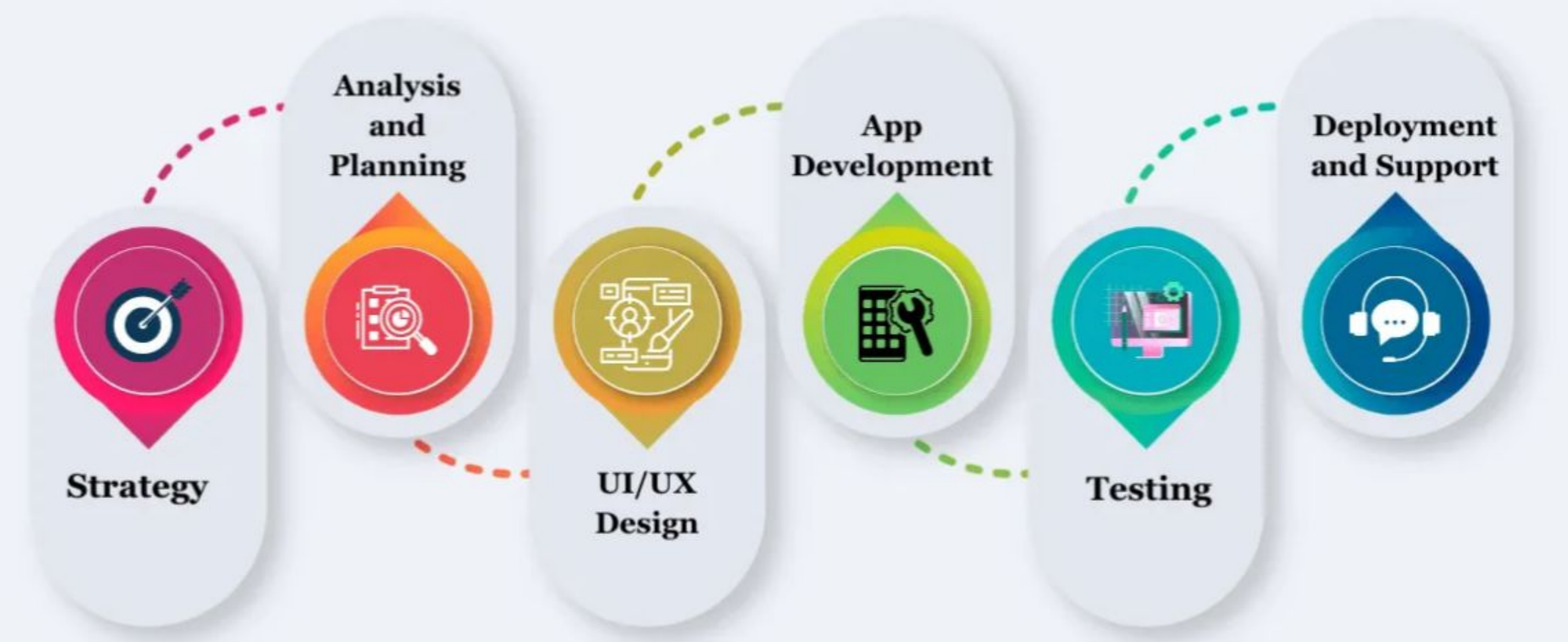
Reminder :

What's the point of studying Software Engineering

Reminder :

To produce a high quality software product via adopting a well-established process model to get tasks and activities well planned.

App Development Process



App Development Process

01

- **Market Research and Strategy**
  - This the most important stage where we :
    1. *Discover an idea*
    2. *Conduct Market Research*
    3. *Validate the idea*
    4. *Draw the strategies to turn the idea into an app* ***for a successful business.***

App Development Process

01

- **Market Research and Strategy**
  - ***Discover an idea***
    - Identify a direct problem that your app can solve it.
      - Observe your daily routine ?
      - Monitor other people at different contexts for problems that you may assist to solve.
    - Follow your passion : Consider creating an app about what interests you most (hobbies, activities .. etc)

App Development Process

01

- **Market Research and Strategy**
  - ***Discover an idea***
    - Penetrate to other markets/Go to the outside world in order to find a niche that you can master. The aim is to explore opportunities that you can exploit.
    - Monitor the market and see what developers are doing

App Development Process

- **Market Research and Strategy**

Sitting behind a computer searching or asking chatGPT for ideas, may not be fruitful. It is highly recommended to go outside and explore the real problems next to you **Yourself**.

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order to find a niche that you can master. The aim is to explore opportunities that you can exploit.

- Monitor the market and see what developers are doing

App Development Process

01

- **Market Research and Strategy**
  - ***Discover an idea***
    - Searching the web ?
      - Analyse what people are searching for
        - *Use keywords Volume Search*
      - Marketplaces where Developers to sell Apps:
        - *Flippa, ..*
      - Websites where developers seek funding
        - *Crunchbase*
        - *AngelList*

App Devel

KD : Keyword Competition Difficulty
Volume : Number of Monthly Searches

01

○ Ho

Keyword ideas for “تمارين النهايات”

42 keywords

Phrase match / Questions

Keyword	KD	Volume ↓	Updated
تمارين النهايات مع الحلول pdf	0	30	15 September
سلسلة تمارين النهايات مع الحلول	N/A	20	
تمارين النهايات للسنة الثانية ثانوي	N/A	20	
تمارين النهايات مع الحلول	N/A	20	
سلسلة تمارين النهايات	N/A	10	
حلول تمارين النهايات للسنة الثانية ثانوي من الكتاب المدرسي	N/A	10	
تمارين النهايات مع الحلول 2 ثانوي	N/A	10	
تمارين النهايات اولى باك علوم تجريبية	N/A	0–10	
حل تمارين النهايات والاستمرارية	N/A	0–10	
تمارين النهايات سنة ثالثة ثانوي	N/A	0–10	

- websites where developers seek funding
  - *Crunchbase*
  - *AngelList*

KD : Keyword Competition Difficulty
Volume : Number of Monthly Searches

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Keyword ideas for “koora”

The first 100 keywords out of 4,072

Phrase match / Questions

Keyword	KD	Volume ↓	Updated
koora	49	979K	1 day
koora live	39	574K	1 day
koora online	23	94K	about 14 hours
koora tv	48	38K	1 day
live koora	37	30K	about 1 hour
koora extra	1	13K	1 day
koora 360	5	8.6K	1 day
el koora	41	6.3K	1 day
koora 4 live	18	6.1K	1 day
koora en ligne	24	5.9K	1 day
yalla koora	Sign up	5.0K	1 day
koora live bein		4.5K	1 day

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eywords Volume Search

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a, ..

where developers seek funding

chbase

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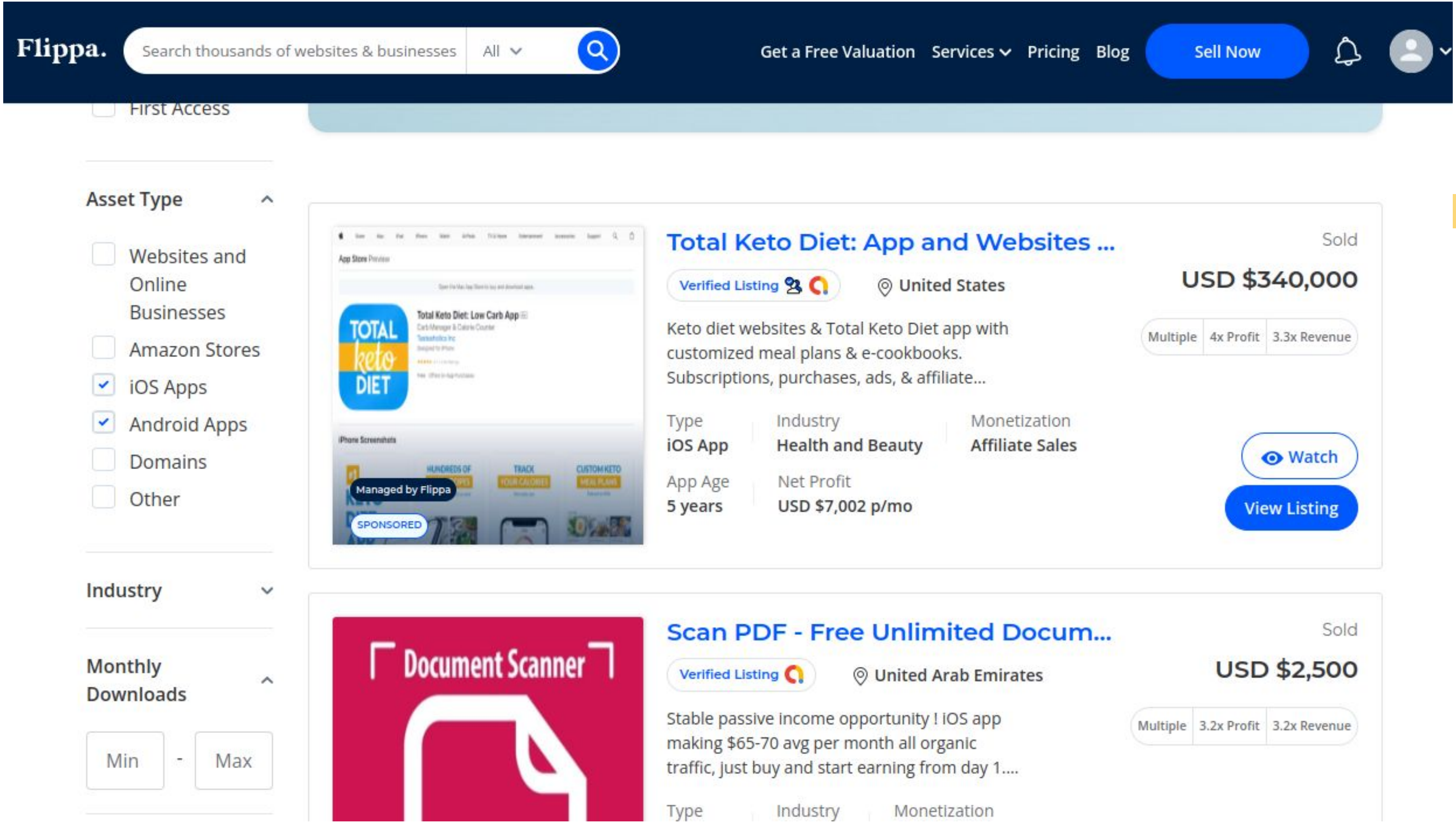
Flippa.
Search thousands of websites & businesses
All
Get a Free Valuation
Services
Pricing
Blog
Sell Now
First Access
Asset Type
Websites and Online Businesses
Amazon Stores
iOS Apps
Android Apps
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Other
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Total Keto Diet: App and Websites ...
Verified Listing
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Sold
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3.3x Revenue
Keto diet websites & Total Keto Diet app with customized meal plans & e-cookbooks. Subscriptions, purchases, ads, & affiliate...
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Health and Beauty
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Affiliate Sales
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Scan PDF - Free Unlimited Docum...
Verified Listing
United Arab Emirates
Sold
USD $2,500
Multiple
3.2x Profit
3.2x Revenue
Stable passive income opportunity ! iOS app making $65-70 avg per month all organic traffic, just buy and start earning from day 1....
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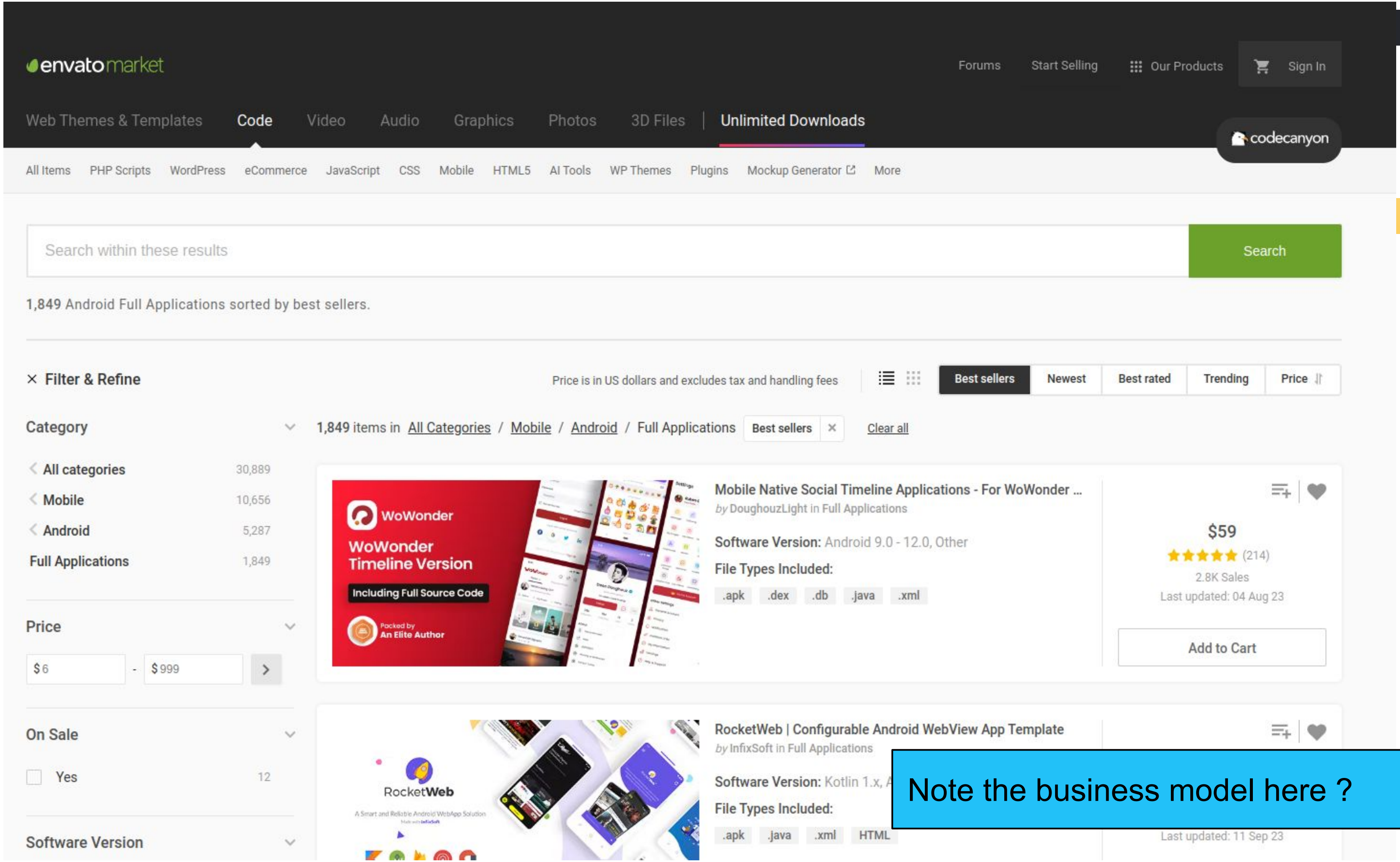
Industry
Monetization
Document Scanner



Note the business model here ?

App Development Process

01

- **Market Research and Strategy**
  - ***Discover an idea***
    - Read reviews for existing solutions to find a gap
    - Browse existing datasets for possible insights to get an idea
    - Explore other apps and see if you can :
      - Improve
      - Localize to your region.
    - ..

App Development Process

01

- **Market Research and Strategy**
  - ***Discover an idea :*** **Does the idea need to be original ?**
    - It's rare now to claim that our ideas are original and none has thought about them. There are **26 million** of software developers (and entrepreneurs) thinking daily on how to create a good app.
    - Look at existing ideas and see how you can revamp them with different settings, business model, different types of audience (Example, Yassir)

App Development Process

01

- **Market Research and Strategy**
  - ***Conduct Market Research*** *:* If you have an idea
    - Conduct intensive market research on whether the idea can be turned into a successful app/business.
    - Survey the competitions and see what gaps and opportunities you can exploit.
    - Collect as much statistical data about the competitions, similar apps as it helps you to gain deeper insights and have a clear vision.

App Development Process

01

- **Market Research and Strategy**
  - ***Idea Validation***
  - = is the process to verify:
    - The assumption that your **targeted audience truly needs** your app to perform the intended task in order to solve an existing problem.
    - The idea can be developed as an App within the available resources (Technology, Funding, HR, Legal....)

App Development Process

01

- **Market Research and Strategy**
  - ***Idea Validation :*** How to validate an idea for an app
    - Defend why you try to solve the problem by writing a good and convincing problem statement answering the following questions:
      - ***What is the problem?***
      - ***Who has the problem?***
      - ***Why is the problem important?***
      - ***How the competitors failed to address the problem***
      - ***How does your app solve the problem?***

App Development Process

01

- **Market Research and Strategy**
  - ***Idea Validation :*** How to validate an idea for an app
    - Market research on :
      - The size of your market (# potential users)
        - Use keywords volume search
        - #downloads can give insight too.
      - The size of the competitions (# apps , # downloads)
    - Assess the technical/financial feasibility
    - Seek feedback from : potential users + experts

App Development Process

01

- **Market Research and Strategy**
  - **Define the strategy** to turn your idea into a successful app via :
    - Define the main objective of your app. “This App would do X and Y”. Use plain and Simple English, Be concise. Example : ***An App to help*** **BAC students to understand the function limits**
    - Identify your potential app users : Bac Students
    - Choose the mobile platform(s) : Android and iOS

App Development Process

01

- **Market Research and Strategy**
  - **Define the strategy** to turn your idea into a successful app via :
    - Define a strategy to
      - Gain more users.
      - Keep the retention rate high
      - Keep your users active/engaged
      - Monetization process or business model for a sustainable app.

App Development Process

02

- **Analysis & Planning**
  - Requirement Engineering is performed to :
    - **For the features to implement:**
      - Capture the functional requirements
      - List potential wish-to-have features
      - Prioritize the functionalities based on their added-value vs. complexity vs. cost
      - List all non-functional requirements including scalability, storage, synching of data, speed,

App Development Process

02

- **Analysis & Planning**
  - Requirement Engineering is performed to :
    - **For the users to implement:**
      - Identify the main target type of users (Kids ? Employees at a given factory? Motorbike drivers ...)
      - Analyse and study their attributes and capabilities:
        - Educational level
        - Preferred language
        - Any accessibility issue

App Development Process

02

- **Analysis & Planning**
  - Requirement Engineering is performed to :
    - **For the hardware and Platform:**
      - Analyze if there is are specific requirements for:
      - Hardware
        - Limited Screen Size ? Special Sensors ? Limited Memory ? Use of Stylus ? Has integrated Keyboard ? Battry Size vs Consumption ?
      - Platform :
        - Special operating system ? integration with legacy system

App Development Process

02

- **Analysis & Planning**
  - Requirement Engineering is performed to :
    - **For the context and environment:**
      - Need to work in outdoor conditions.
      - Remote area with unstable internet connection
      - Users always on the move, need mostly voice-interaction instead…
      - Lighting is dark ? need better use of colors/contrast

App Development Process

02

- **Analysis & Planning**
  - After collecting the raw requirements :
    - Requirements are analysed for :
      - Inconsistencies, conflict, missing requirements, feasibility, ambiguities, overlap…
    - The requirements are documented.
    - Requirements are verified and validated with the stakeholders.
    - Software Requirements Specification (SRS) document is produced

App Development Process

02

- **Analysis & Planning**
  - A product roadmap/Timeline is prepared showing clearly the dates and milestones of the project. This includes:
    - *Release of the prototype/Wireframe*
    - *Branding and Identity*
    - *Backend Architecture*
    - *Licensing and Administrative Procedures*
    - *Hiring & Human Resources*
    - *Publishing to App Stores*
    - *MVP Release*
    - *Dev Sprints (High Level)*
    - *Marketing events*
    - *etc..*

App Development Process

02

- **Analysis & Planning**
  - Planning for Marketing and Promotion events
  - Budget Planning for the project needs to be done.
  - How much financial resources are need to pay for :
    - *Servers*
    - *Services and APIs*
    - *Designers*
    - *Copyrights*
    - *Staff…*

App Development Process

03

- **UI & UX Design**
  - For Mobile Apps, the success of the app is largely dependent on the perceived :
    - User experience : …
    - Usability aspect : ..
    - Artistic/Esthetic Look : ..
  - (Of course, in addition to the utility)

App Development Process

03

- **UI & UX Design**
  - The goal of design is to design an easy to use app creating an excellent user experience to navigate easily between screens to perform the intended task.
  - The design phase is extremely important, critical and time-consuming (Can be **very** expensive when outsourced)
  - As opposed to Enterprise Software Systems where performance, security, scalability…Mobile Apps are user-centric. They revolve around the user.

App Development Process

03

- **UI & UX Design**
  - Designers will deliver:
    - *Wireframes*
    - *Flow of Navigation between Screens*
    - *Prototypes*
    - *Mockups*
    - *Coloring Theme*
    - *Icons*
    - *Fonts*
    - *Images*
    - …

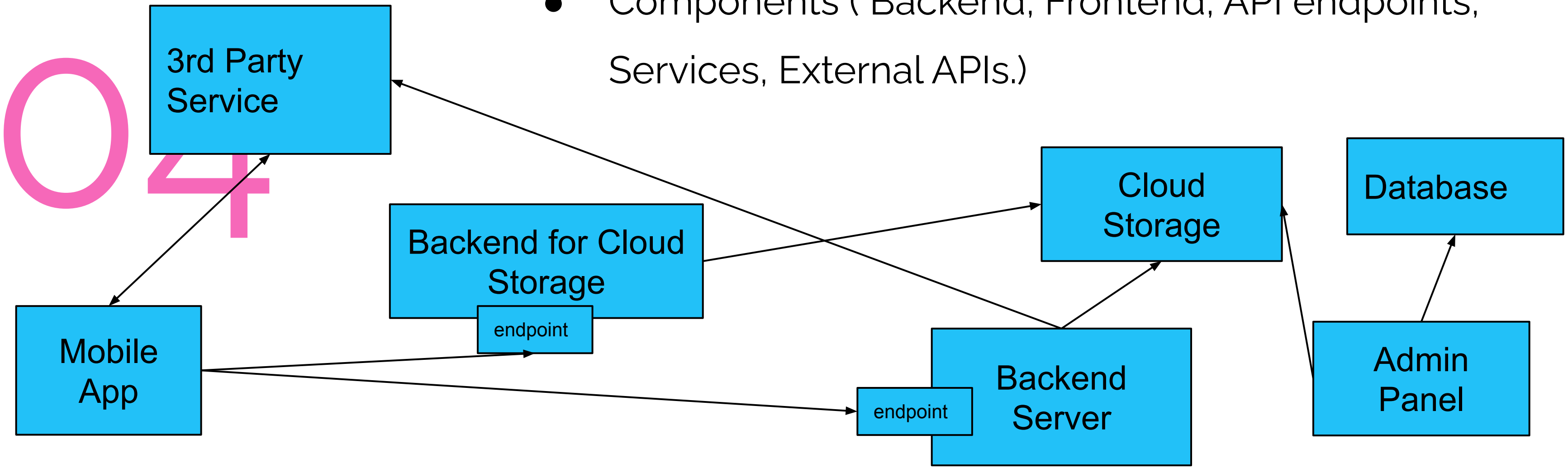
App Development Process

04

- **Development**:
    1. Design the architecture of the software
    2. Select the technology stack
    3. Implementation and Integration

App Development Process

- **Development**
    1. Design the architecture of the software
        - Components (Backend, Frontend, API endpoints, Services, External APIs.)



App Development Process

04

- **Development**
  1. Design the architecture of the software
     - Low-Level Design of each component : UML diagrams can be utilised to show :
       - Classes, Attributes and Relationships
       - Algorithms/Business logic for certain functionalities

App Development Process

04

- **Development**
    1. Design the architecture of the software
        - Communication Design : to describe the data flow and connection between various components.
            - Using RPC
            - Rest APIs
            - Queue-Based Communications.

App Development Process

- **Development**

  1. Design the architecture of the software

  - Database and Data Design :
    - Database Structures and Technologies
      - Local Database
      - Remote Database
      - 3rd Party Database
    - Schema/format for data (JSON, XML, CSV)

04

App Development Process

04

- **Development**

    2. Select the technology stack:

        - Platforms :
            - Operating System:
                - Android
                - iOS
                - Both
            - Hardware :
                - Phone, vs TV vs. Smart Watch vs …

App Development Process

- **Development**

2. Select the technology stack:

- Programming languages :
  - Kotlin : Becoming the preferred language for Android. Can be potentially used for iOS , there is a trend to use it (Netflix..).
  - Java : For android native apps + backend development + Web
  - Python : mainly for backend development + Web
  - Dart : The programming language for the flutter framework.
  - Swift : is the primary programming language for iOS app development.
  - Objective-C : Still popular for iOS, being slowly replaced by Swift.
  - JavaScript : mostly for web-based frameworks

04

App Development Process

- **Development**

  2. Select the technology stack:

     - Frameworks :
       - Flutter : Cross-platform framework developed for Google for building android and iOS apps
       - React Native : Developed by Facebook for developing iOS and Android native apps using JavaScript Technologies.
       - Xamarin : Cross-platform which uses C# and . iOS, Android and Windows. Owned by Microsoft.
       - Cordova (Ex-PhoneGap) : For building hybrid apps using web technologies.
       - Ionic : For building hybrid apps using web technologies

04

App Development Process

- **Development**

2. Select the technology stack:

- Database Engines :
  - SQLITE : supports both iOS and Android, Embedded, Serverless relational databases.
  - Realm : Supports both iOS and android.
  - Paid systems (Couchbase mobile..)
  - Local file system

04

App Development Process

- **Development**

    2. Select the technology stack:

        - Backend Services
            - Java Spring Boot
            - Django
            - Flask
            - Java Spark
            - Laravel

04

App Development Process

- **Development**

  2. Select the technology stack:

     - External APIs and Services
       - Firebase
       - Backendless
       - Kinvey
       - Appery.io

04

App Development Process

04

- **Development**

  3. Implementation and Integration

     - Development of various components, services..
     - Integration and connecting components with:
       - Each other,
       - External APIs and libraries

App Development Process

05

- **Testing & Quality Assurance**
  - The testing phase is to ensure that the app once released:
    - It meets the highest quality standards
    - Users would not complain about missing features, non-working functionalities, bugs, distorted display....
    - Reduce maintenance and frequent updates…

App Development Process

05

- **Testing & Quality Assurance**
  - The testing activity includes :
    - UX and Usability testing
    - Functionalities Testing + Unit Testing
    - Integration Testing
    - Performance Testing (Includes also scalability, speed..)
    - Security Testing
    - Device and Platform Testing
  - Testing can be performed either in automated or manual way

App Development Process

06

- **Deployment & Maintenance**
  - Either
    - Publish within the App Stores.
    - Deploy privately at the designated devices.
  - Monitor constantly feedback/reviews/ratings and take the necessary measures (Address feedback, reply to questions, submit bugs to your internal issue-tracking system..)

App Development Process

- **Analysis, Develop, Test and Publish :**
  - It is an iterative process
  - With every incremental update you make to your application, it is crucial to test your app and see if it stands up to expectations or not
  - Testing an already released app, requires:
    - *Testing all functionalities (including the ones not-affected by)*
    - *Testing with existing data*
    - *Testing the migration if there is*

Usability & UX : Design Principles

- **Usability**
  - is defined as the extent to which a product can be easily used by specified users to achieve certain goals with effectiveness, efficiency and satisfaction.
  - Simply : how easy to use a software product to perform a task with few clicks, speed and better learnability rate…
  - Software systems are valued on the basis of its graphical interface and the related power of communication and expression for the implemented functionalities

Usability & UX : Design Principles

- **Aspects of Usability**
  - There are five aspects that constitute the usability :
    1. *Learnability : how easy to learn how to use the software*
    2. *Efficiency : how fast to perform a given task*
    3. *Memorability : Can I remember later on doing the previously done task ? and after a long time ? can I still remember ?*
    4. *Errors : Few errors, for the user's errors, I can undo them..*
    5. *Satisfaction : I feel satisfied, i feel having control of the*

Usability & UX : Design Principles

- **User Experience**
  - Goes beyond the product interface, covering the entire experience with a product: how good the product was at helping them complete a job and did it work as expected and how they felt while using it
  - User experience (UX) can be simplified into four elements:
    - ***Usability*** *– Was it easy to complete a task?*
    - ***Adoptability*** *– Do I want to repeat the experience?*
    - ***Desirability*** *– Was using the product fun and engaging?*
    - ***Value*** *– Is the product providing value – does it help me complete my task?*

Usability & UX : Design Principles

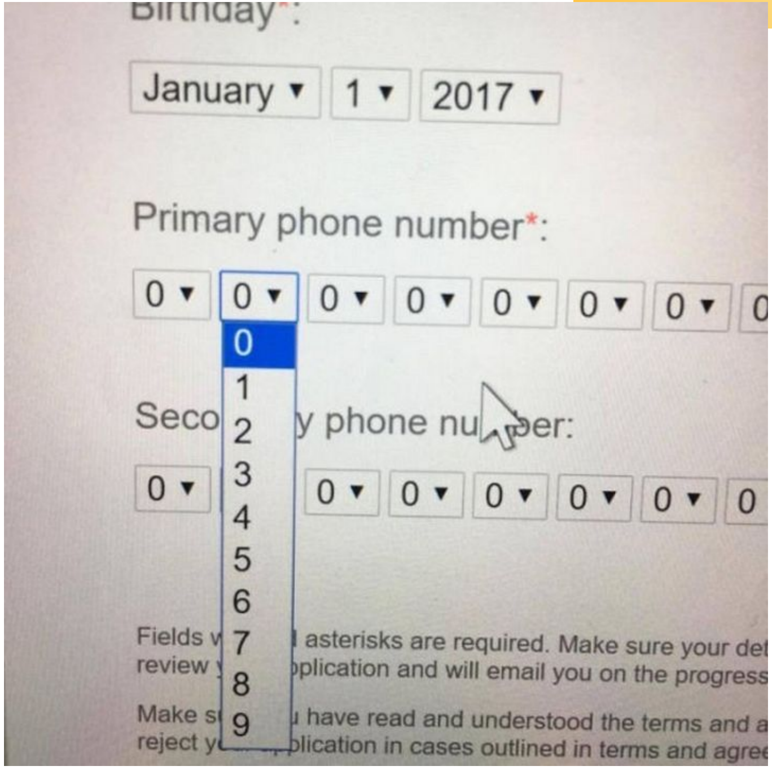
- **Design Principles : Nielsen's 10 Usability Heuristics (1994)**

    1. *Visibility of system status*
    2. *Match between system and the real world*
    3. *User control and freedom*
    4. *Consistency and standards*
    5. *Help users recognize, diagnose and recover from errors*
    6. *Error prevention*
    7. *Recognition rather than recall*
    8. *Flexibility and efficiency of use*
    9. *Aesthetic and minimalist design*
    10. *Help and documentation*

Usability & UX : Design Principles

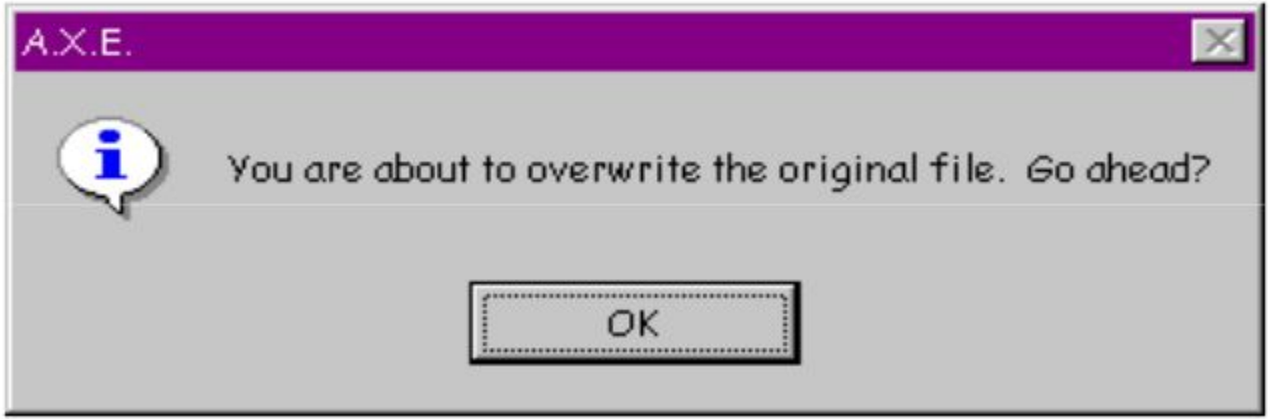
Is this the efficient way to ask the user to provide their phone number ?

Is this the way we write phone number in in the real world ?



Usability & UX : Design Principles

Is the user feeling he is in control of their app ?



Usability & UX : Design Principles

In terms of norms and standards, we ask people with a negative sentence against the default action ?

Analogy :

Someone is invited to a table to eat and you tell **“Ok, not to eat dinner ?”**

Usability & UX : Design Principles

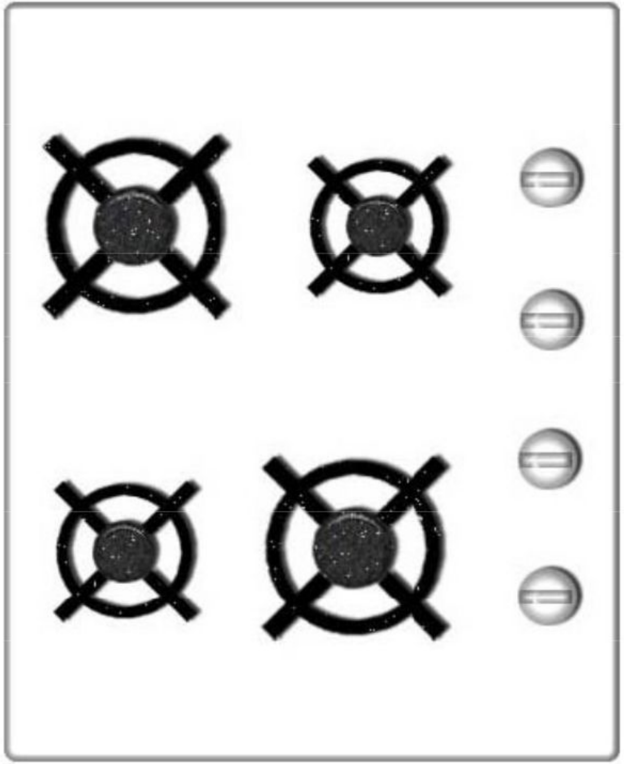
Consistency of the message ?

Usability & UX : Design Principles

How can I switch on the big hob at the top ?

I can learn now,

but , Memorability ? Can I recall after a long time ?



Usability & UX : Design Principles

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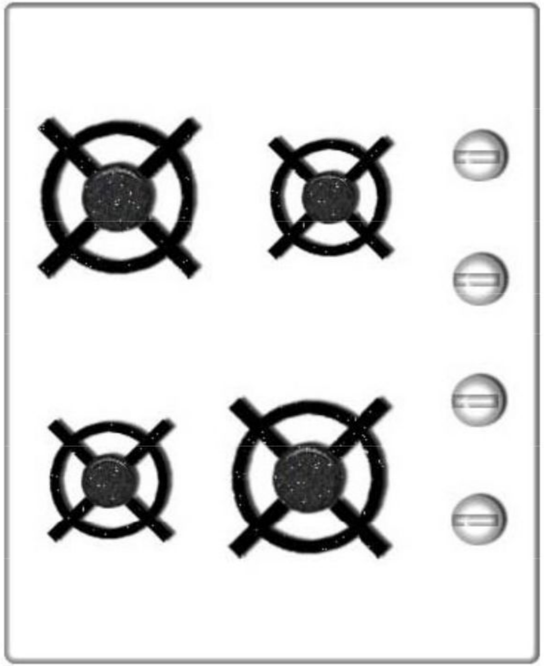
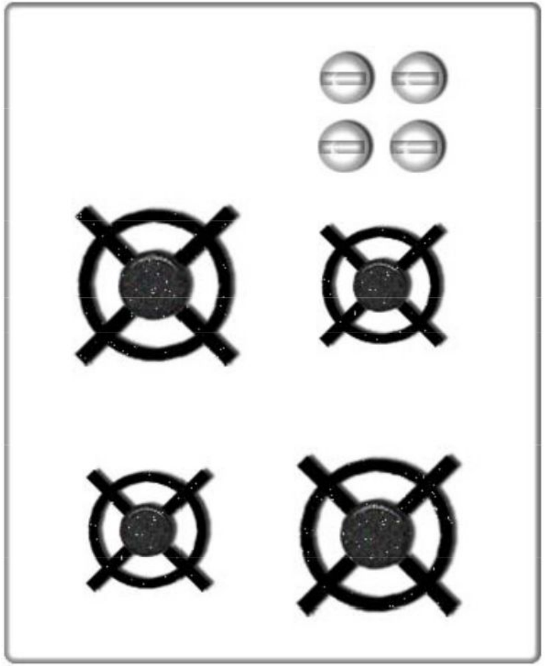
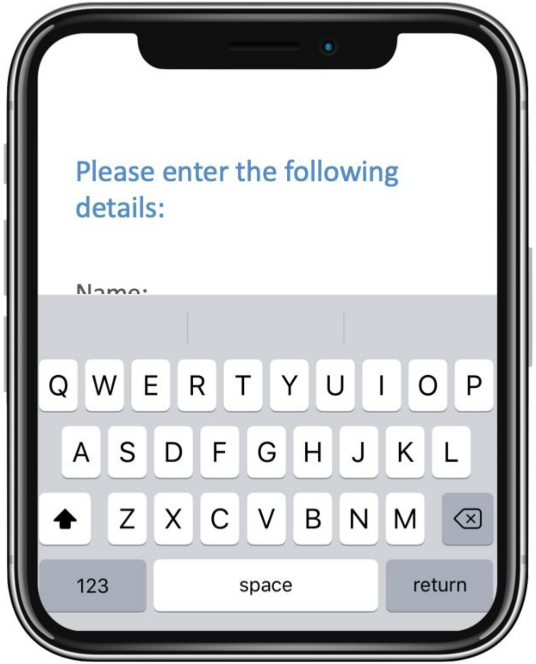
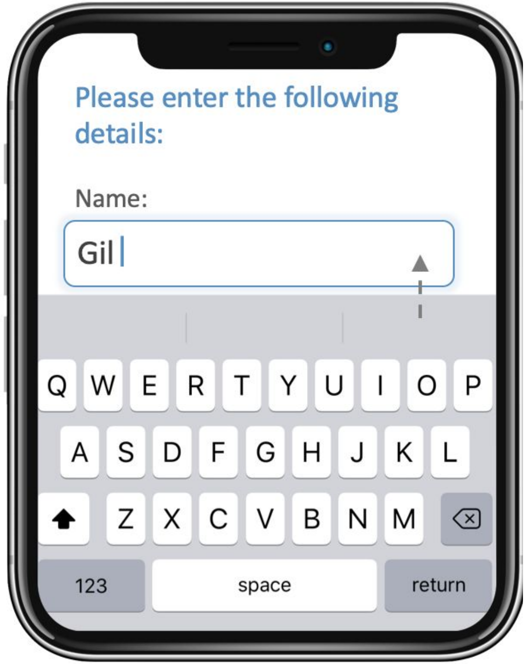




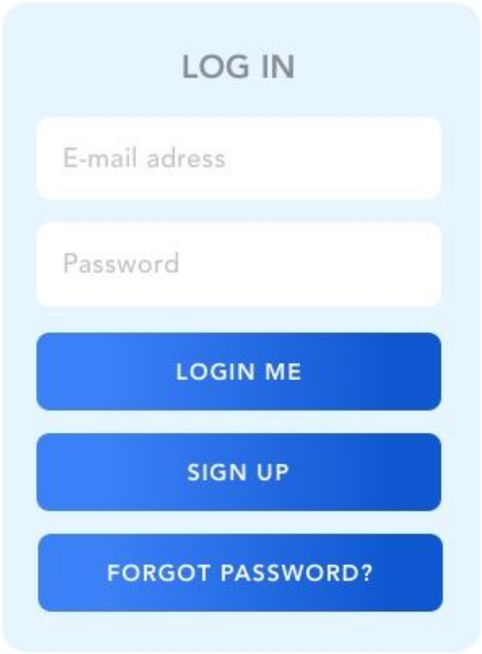
Usability & UX : Design Principles





Usability & UX : Design Principles

As humans, do do we truly read ?



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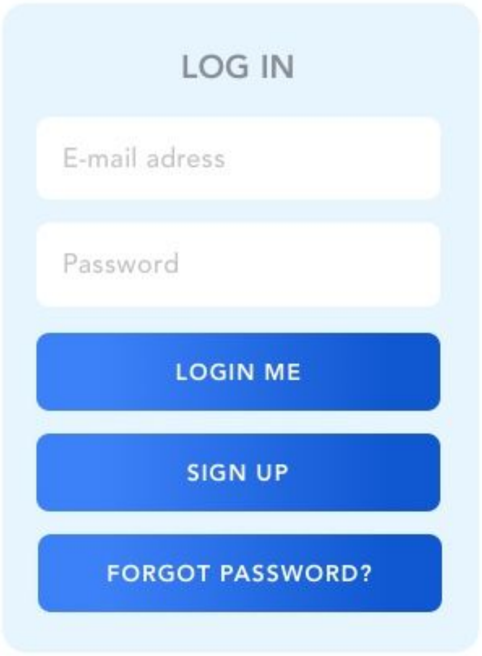

Usability & UX : Design Principles

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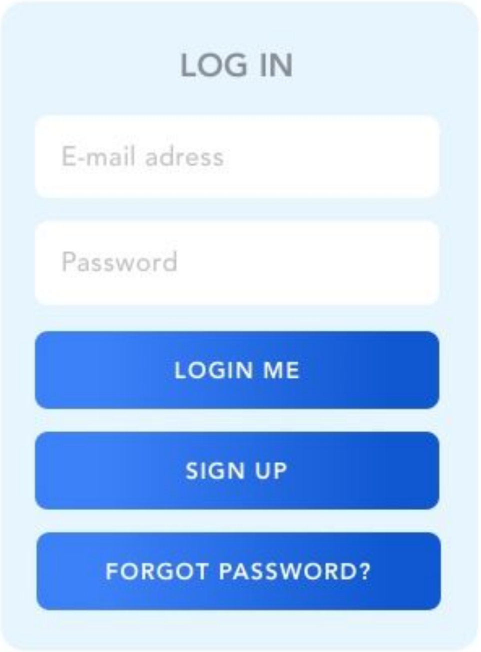
Le rsete peut êrte dnas un dsérorde ttoal et vuos puoevz tujoruos lrie snas porlblème.

C'est prace que le creaveu hmauin ne lit pas chuaqe ltetre elle-mmêe, mias le mot cmome un tuot.



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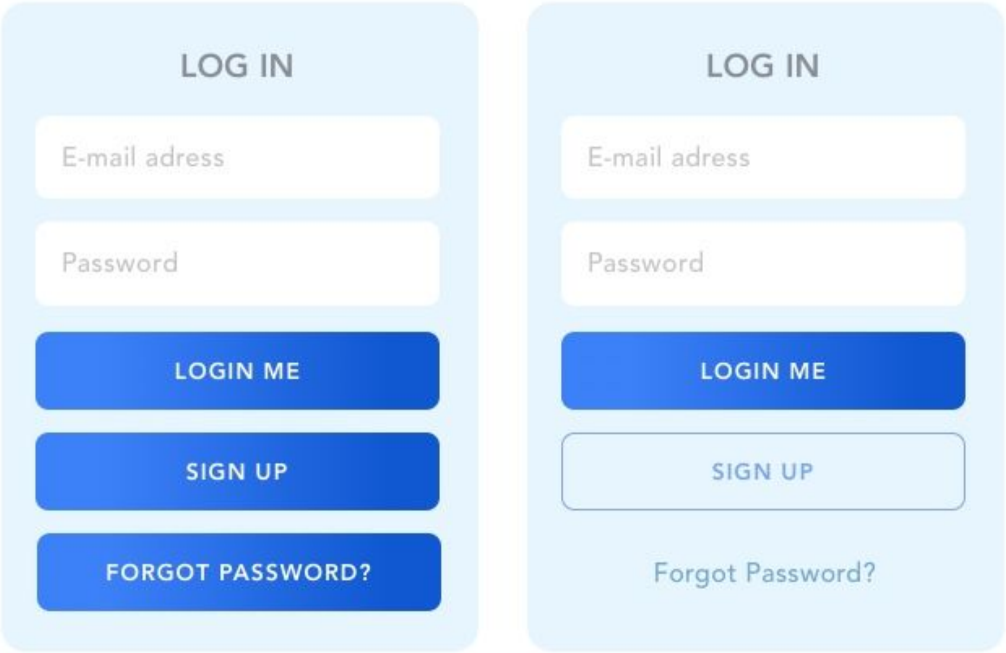

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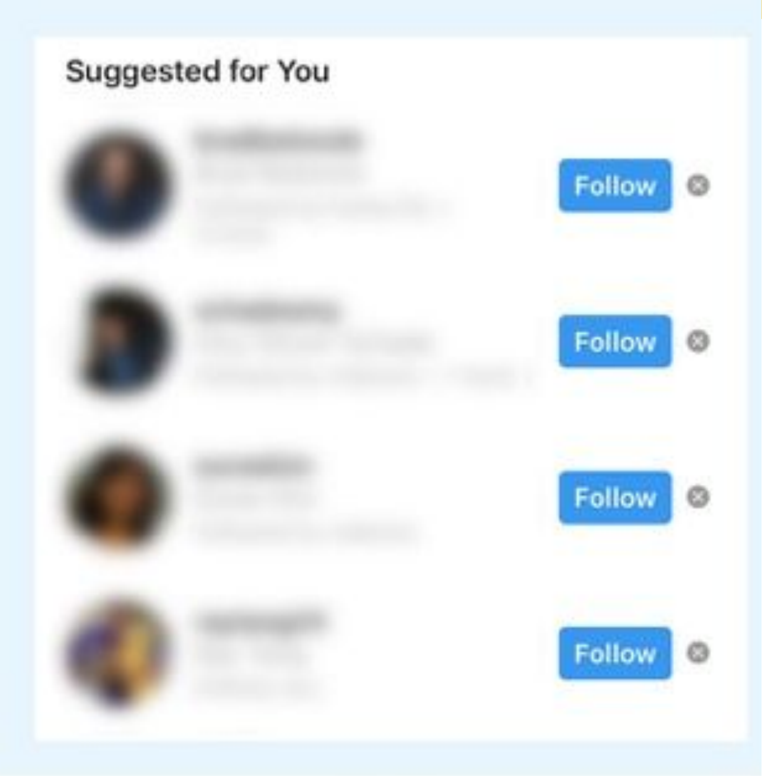


Usability & UX : Design Principles

- **Recommended Design Principles for Mobile Apps:**
  1. **Require minimal intervention** : fewer steps to solve a problem, avoid unnecessary screens, form fields that can slow the users and even clutter the screen
     - Allow Flexible input
     - Provide smart defaults (Default country ..)
     - Support direct manipulation, shortcuts of menu actions using hand gesture (zooming an image..)

Usability & UX : Design Principles

- **Recommended Design Principles for Mobile Apps:**

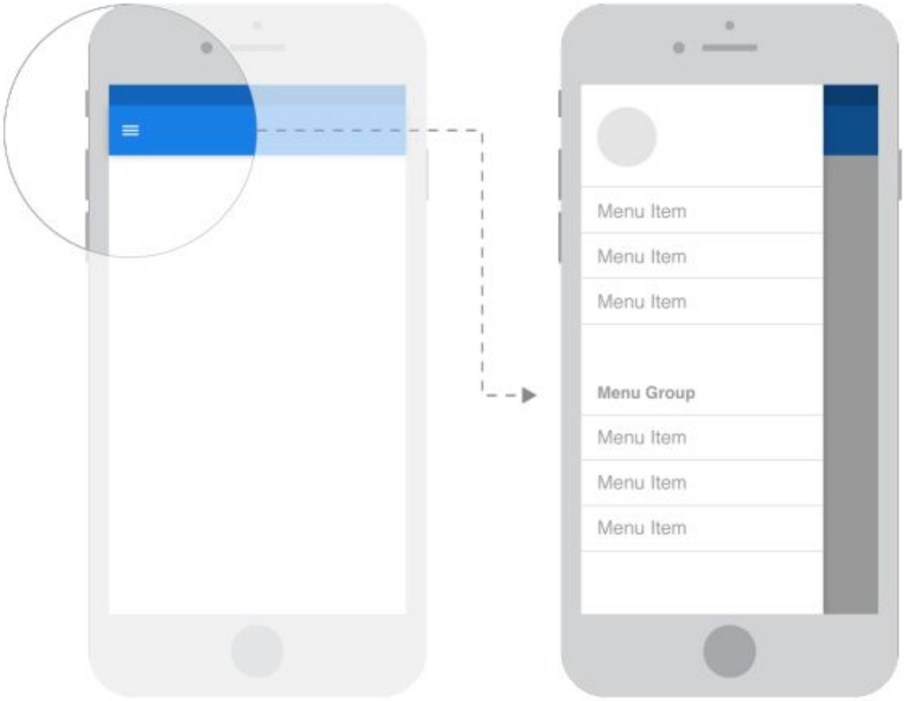
  2. **Inform users in relevant terms** : Users want to know what is happening in meaningful and simple language
     - Make users feel in control:
     - Communicate in the user's terms
  3. **Don't waste the user's time :**
     - Performance
     - Reduce interruptions : Avoid Modal dialog, alerts…

Usability & UX : Design Principles

- **Recommended Design Principles for Mobile Apps:**
  4. **Avoid mistakes** : Users always and always make errors and mistakes. Your product should **never** make them feel stupid or harm them
     - Make it impossible to use it incorrectly
     - Offer the possibility to undo or correct the errors

Design Patterns for Mobile Apps

- **Navigation :**
  - **Sliding drawer**
  - Upper area tabs
  - Bottom menu
  - Back button
  - Floating buttons



Design Patterns for Mobile Apps

- **Navigation :**
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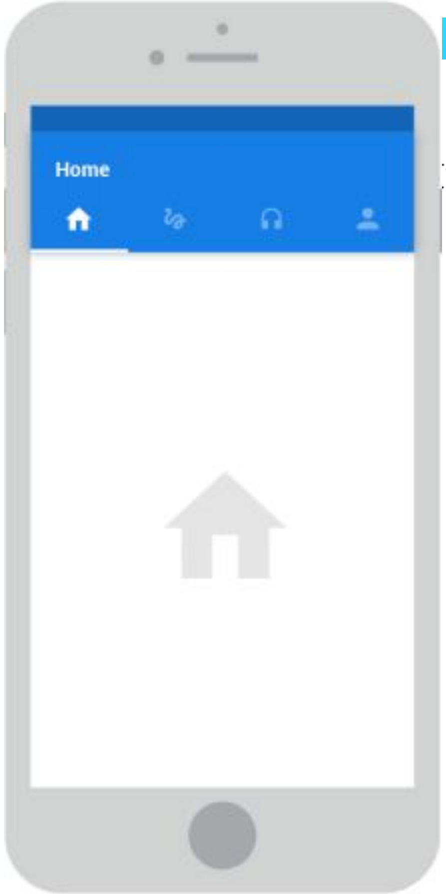

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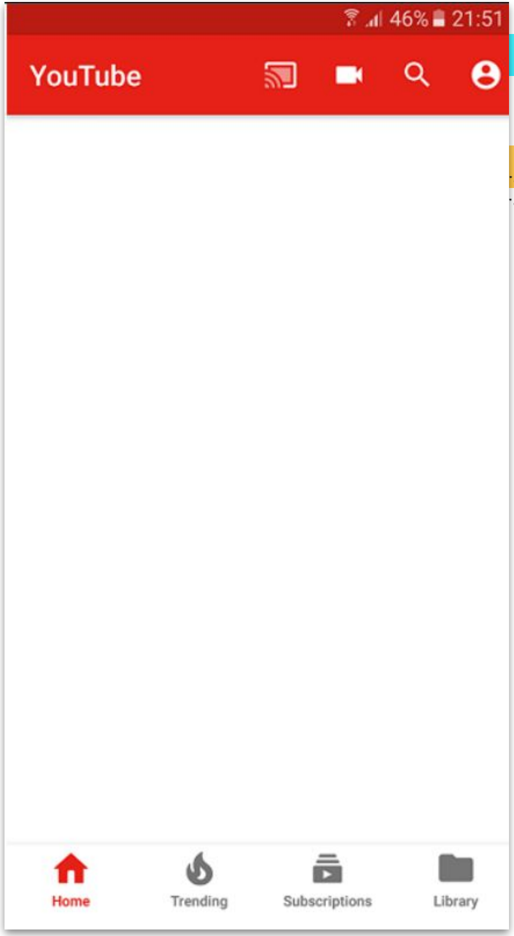

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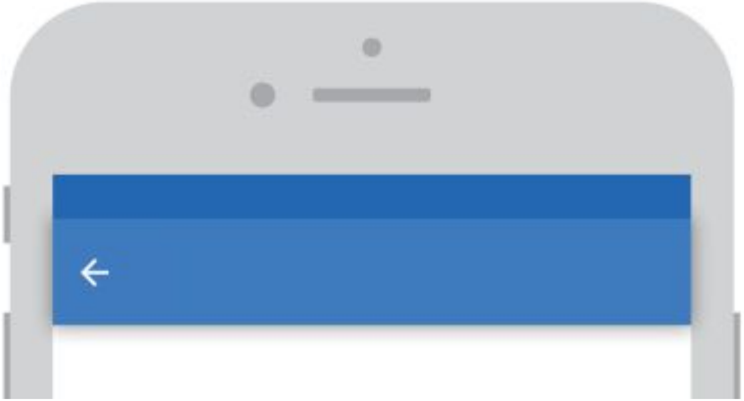

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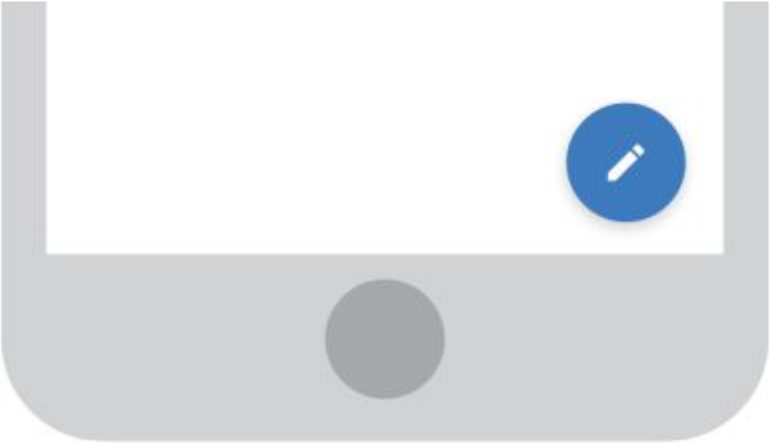

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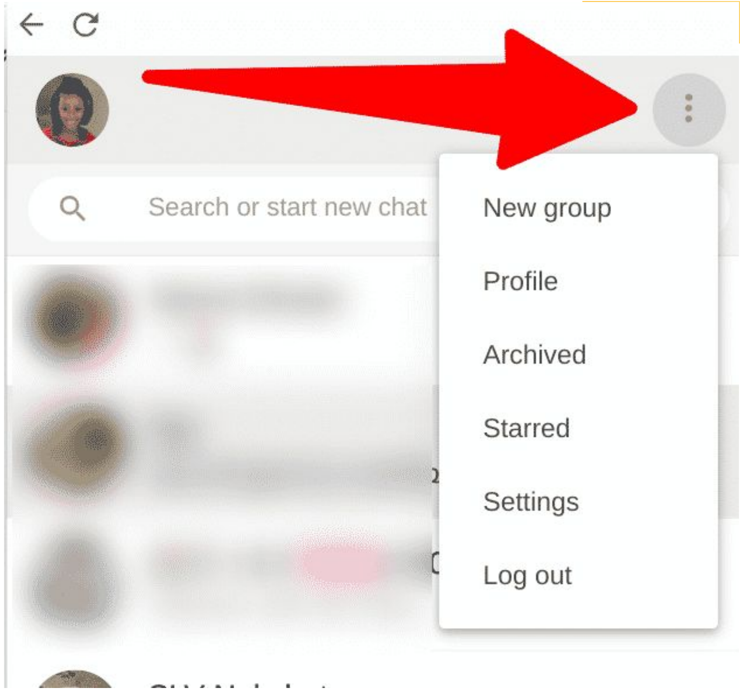
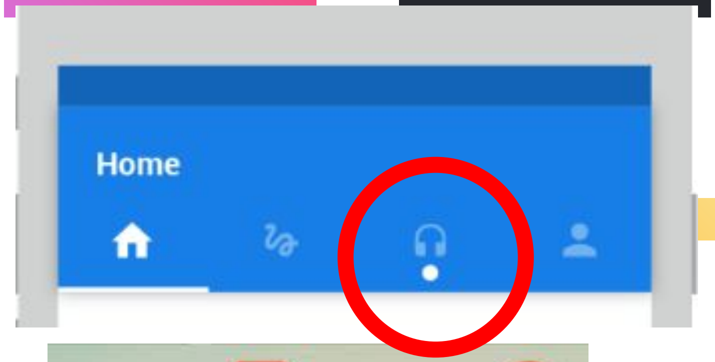


Design Patterns for Mobile Apps

- **Notifications :**
  - Movement
  - Circle hint
  - Count notification app icon
  - Icons on Android Status bar



Design Patterns for Mobile Apps

- **Behavior Patterns :**
  - **Progressive Disclosure**
  - Lazy Login
  - Swipe down to refresh

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Business Models for Mobile Apps

- **What Business Model :**
  - A business model is a “value capture framework” via which you monetize your product or service based on the value it creates for your customers.
    - *Hence, it is value-based and NOT cost-based*
  - There is no one universally right business model, as it depends on your specific situation

Business Models for Mobile Apps

- **Business Model : Up-Front Charge Model**
    - A customer pays a large one-time up-front amount of money to obtain a product/service
    - Typically, the customer can also secure ongoing upgrades or maintenance of the product for a recurring fee
    - The model can serve in a large up-front infusion of cash to your business, but might impact your ability to secure a good enough recurring revenue stream.

Business Models for Mobile Apps

- **Business Model : Usage-Based Model**
  - The usage-based model is similar to how electric and water utilities are metered (i.e., pay-as-you-go a service fee)
    - *E.g., Cloud Computing services like Amazon EC2*
  - The service fee is typically constant and NO up-front (or base or subscription) fee is incurred
  - The model provides control to customers over their expenses because they only pay for the amount of resources they use (rather than paying for extra capacity they do not use)

Business Models for Mobile Apps

- **Business Model : The Subscription or Leasing Model**
  - A customer pays a subscription fee at the end of every predetermined time period (e.g., bi-weekly, monthly, annually, etc.) for obtaining your service or product E.g., Netflix
  - Typically, you can extract higher payments over shorter periods of time (e.g., monthly vs. yearly)
  - The model provides flexibility to customers (assuming they can unsubscribe at anytime) with predictable, fixed payments for businesses over agreed-upon periods of time

Business Models for Mobile Apps

- **Business Model : The Upsell Model**
    - You can sell your product at a very low margin (or even free), but increase your overall margin via selling add-on products and/or charging for necessary future services
        - *E.g., Consumer electronics (e.g., cameras) and automobiles (add-ons include warranty extensions, accessories, etc.)*
    - Attractive for customers
        - *Initially, they might not pay much*
    - Profitable for businesses
        - *They usually incur large margins on add-ons*

Business Models for Mobile Apps

- **Business Model : Freemium Model**
  - A customer pays zero money for a basic functionality of your product, but pays for obtaining premium features.
  - Many people can try your product
    - *However, will these people (they are not customers until they pay) pay for the extra features available in your product?*
  - Caveat: If people do not pay for your extra features, you do not have a business (recall the solo condition for having a business)

Business Models for Mobile Apps

- **Business Model : Advertising Model**
  - You can make your product free, but monetize your ability to attract and retain a desirable demographic via providing Ads for third parties who want access to them (E.g., Google's AdMob)
  - Appealing to users and third parties, especially that Ads are seamless (no banners!) and targeted
  - Caveat: many startups have fallen substantially short when they relied solely on Ads

Business Models for Mobile Apps

- **Business Model : Franchise or White-label Model..**
  - If you own a good product with a good brand, you may let other business conduct business using your brand/app with a compliance to certain procedures...
    - The franchisee will pay a royalty fee or commissions from their sales/profit.

Business Models for Mobile Apps

- **Business Model : Franchise or White-label Model..**
  - If you own a good product but not necessarily a good brand, you may let other business conduct business using your brand/app but their brand instead of your brand without the need to comply to regulations imposed with the franchise model.
    - Monetization can be achieved via a subscription fee, or commission based on their sales.

Business Models for Mobile Apps

- **Business Model : The "Parking Meter" Model**
  - Parking meters charge low hourly parking rates (e.g., $0.25)
    - This seems to defy the logic of having large and expensive parking meters
  - However, the city charges high for parking tickets that become even higher if someone does not pay in an X (e.g., 10) number of days
    - No wonder cities have so many parking enforcement people!
    - Is not this aspect of the model akin to the one used by credit/debit card companies against defaulters?
  - Caveat: loyal customers might become alienated by late fees (discovered by Blockbuster and precluded by Netflix)

Business Models for Mobile Apps

There are more and more…

Resources

- Pablo Perea, Pau Giner. “UX Design for Mobile”, 2017 (Good Book on Mobile App Design)
- https://www.youtube.com/watch?v=u7iUoxqKaKU&ab_channel=GoogleforDevelopers
- https://www.youtube.com/watch?v=1bzWw2uZZuA&list=PLlgPXNRcnX3HZDuZIMidALLtgMA7sRUGR&ab_channel=InteractionDesignFoundation%E2%80%93UXDesignCourses
- https://www.ihci-conf.org/wp-content/uploads/2021/07/04_202105L012_Lupanda.pdf
- https://www.agileinfoways.com/blogs/mobile-app-development-process/
- https://www.saabsoft.com/blog/mobile-app-development-process/
- https://improvement.stanford.edu/resources/usability-principles
- https://userpilot.com/blog/usability-vs-user-experience/